

ELEVATED LEAD EXPOSURE IN ROMAN OCCUPANTS OF LONDINIUM: NEW EVIDENCE FROM THE ARCHAEOLOGICAL RECORD*

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The evolution of metallurgical practices over several millennia has resulted in human exposures to a diverse variety of toxic elements. These exposures were not just confined to the metalworkers, but in many cases broadly impacted the larger community and society. The role of lead in Roman civilization is particularly fascinating, where the available evidence of lead use in a myriad of applications by Roman civilization suggests the potential for elevated exposures. The present paper addresses the hypothesis that because of these practices, Roman occupants of ancient Londinium (43–410 AD) were exposed to levels of lead that far exceeded exposures in the preceding populations of Iron Age Britain (700 BC–43 AD). The elemental content and stable lead and strontium isotopic signatures of 30 femora from three Londinium cemeteries and radiocarbon-dated representative burials were examined and the results then compared with 70 femora dating to the pre-Roman Iron Age from the well-known Iron Age hillfort of Danebury, Hampshire, and from three cemeteries in East Yorkshire. Extensive efforts were directed at the minimization of sediment and soil contamination, the assessment of contamination from burial artefacts and a careful review of the burial context of specimens. Data for lead and 20 additional trace and major elements were obtained by magnetic sector inductively coupled plasma mass spectrometry (ICPMS), after acid dissolution of pre-cleaned bone cores. Lead isotope ratios in the bone core digests (and burial soils) were measured by multi-collector ICPMS. It was found that concentrations of lead in the Roman/Londinium-era

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femora were > 70-fold greater than those from pre-Roman populations. This was confirmed by femoral data obtained from the analysis of a pre-Roman burial from the early first century BC found adjacent to the Southern Cemetery. Several major and trace element metrics for diagenesis demonstrate that levels of lead in the bone are robust indicators of human exposure during life. Lead isotope data for the Roman population are consistent with previously identified widespread lead pollution. The pre-Roman populations contain lead isotopic compositions consistent with highly localized, minor sources of lead. Together, these data provide unequivocal confirmation that the Romans in Londinium were exposed to elevated lead levels. Elevated blood lead levels would have resulted, negatively affecting their health and possibly contributing to declining birth rates.

KEYWORDS: LEAD, LONDINIUM, ROMAN BRITAIN, IRON AGE

INTRODUCTION

Lead (Pb) pollution due to Roman mining and metallurgy occurs throughout the Iberian plateau (Hillman *et al.*, 2017), the Balkan region (Longman *et al.*, 2018), and within the Bay of Naples (Delile *et al.*, 2016). Furthermore, increased atmospheric deposition of Pb during the Roman period is documented in peat bogs using Pb concentrations and isotope systematics of peat cores throughout Europe (e.g., Shotyk *et al.*, 1998; Allan *et al.*, 2018). Many Roman artefacts are either made of pure, native Pb or contain weight percentages of Pb in bronze and other metal alloys (Dungworth 1997, *passim*). Direct ingestion of Pb-containing compounds in Roman culture is also well documented, for example, the use of Pb acetate as a sweetener in foods and wine (Gilfillan, 1965; Lessler, 1988). In addition, secondary ingestion of Pb due to extensive use of Pb pipes for plumbing has also been hypothesized (Delile *et al.*, 2017). This wide use and intake of Pb is likely to have elevated Pb in Roman human populations. Geochemical analyses showing high Pb levels in Roman-period human bones support this hypothesis (Waldron *et al.*, 1976; Patterson *et al.*, 1987), but the entire concept of Pb poisoning among the Romans is not universally accepted (Scarborough, 1984; Retief and Cilliers, 2005).

In the last decade, a concerted effort has been made to test the original hypothesis of Pb exposure in the Roman Empire put forward by earlier work. Pb isotope and concentration data in tooth enamel point to an increase in Pb content beginning in the late Iron Age, with a particularly elevated Pb level in Roman teeth from the first to fourth centuries AD, which corresponds to a relatively homogeneous, anthropogenically derived Pb isotope signal (Montgomery *et al.*, 2005, 2010). A recent comprehensive compilation of Pb in tooth enamel by Kamenov and Gulson (2014) also documents this elevated Pb content in humans during the Roman period. However, a critical piece is missing from any tooth enamel study; because formation of tooth enamel ceases upon the emergence of the tooth from the jaw, enamel only records the early life conditions of an individual, and does not provide any quantitative information on the degree of exposure over a person's lifetime. In contrast, bones regenerate during the lifetime of an individual, allowing Pb to accumulate in human bones. Thus, Pb in bone provides a means to quantify the true metric of exposure: blood Pb levels. The physiological half-life of Pb in human finger bones has been measured at 16 years (Nilsson *et al.*, 1991), and the half-life of Pb in the human skeleton in general increases as exposure increases (Brito *et al.*, 2014). The established relationship between blood Pb levels and bone Pb concentrations (e.g., Flemming *et al.*, 1997) suggest that measuring the Pb content of archaeological bone could provide a quantitative analysis of Pb exposure in ancient populations.

Owing to the potential for use of trace element geochemistry of archaeological bone as a metric for heavy metal exposure in ancient populations, accurate analyses of trace metal contents

in bone are highly desired (Brown and Blakely, 1985; Shafer *et al.*, 2008). However, whether or not Pb patterns are a product of lifestyle or *post mortem* environmental alteration has been a concern (Patterson *et al.*, 1987). The burial environment determines a bone's susceptibility to diagenetic processes that change a skeleton's original chemical composition (e.g., Ericson *et al.*, 1991). While some post-burial alteration is unavoidable, technological innovations and new methods of bone processing have been successful in managing the effects of post-burial modification (Shafer *et al.*, 2008). Furthermore, recent work in bone diagenesis research has shown that different elements are taken up into bone at different rates, and some elements can even be removed from bone during burial (Krajcarz, 2019). Because diagenetic alteration is unavoidable, a detailed assessment is required to extract information about *in vivo* exposure for any archaeological population.

The present study uses detailed, high-precision trace element and isotopic analyses to evaluate Pb status and then the origins of elevated Pb levels in Roman skeletons from the British settlement of Londinium. We measured trace elements in femora from three burial sites in Londinium, as well as in specimens from pre-Roman Iron Age burials from Danebury, Hampshire, and from three cemetery sites in East Yorkshire. Radiocarbon dating was conducted to confirm the age of the specimens, and strontium (Sr) isotopes were measured to determine potential geographic origin. Pb isotopes were measured to determine the probable origins for the Pb. Together, these data enabled it to be determined whether the Roman occupants of Londinium were exposed to higher levels of Pb than pre-Roman, native inhabitants of Britain, and to evaluate the sources of Pb exposure, and to gain an insight into the cultural habits causing increased Pb exposure in Londinium.

MATERIALS AND METHODS

Femora sourcing and provenance

Femora were obtained from both Roman and pre-Roman burial sites in Britain. Roman burial sites included the Eastern, Western and Southern cemeteries of Londinium (Hall, 1996), and the pre-Roman Iron Age burial sites include the well-described Danebury hillfort site (Cunliffe and Poole, 1991; Cunliffe, 1995) and three cemeteries from East Yorkshire (Stead, 1991; Jay & Richards, 2013; Peck, 2013). Accelerator mass spectrometry (AMS) ^{14}C dating was performed by Stafford on nine femora from the East Yorkshire cemeteries (two from Burton Fleming, two from Kirkburn and five from Rudston) and used collagen extracted by published methods (Stafford *et al.*, 1991; Stafford, 2014). A total of 12 samples from the Londinium collections (four from each cemetery) and six specimens from the Danebury hillfort site were submitted to the Oxford Radiocarbon Accelerator Unit (ORAU) for radiocarbon dating. The Danebury results were reported previously (Cunliffe *et al.* 2015) and are summarized with the results. All the ^{14}C measurements in Oxford were made on collagen by ORAU (Brock, Higham, Ditchfield, & Bronk Ramsey, 2010).

Londinium specimens

Romans in Londinium maintained four cemeteries located on the main roads exiting the city. The largest (16 ha) Eastern site was used for burials throughout the Roman period (Hall, 1996) and flanked the road to Colchester that exited Londinium at Aldgate. In the west, burials were located in the area around the modern Holborn and Smithfield areas near Newgate. South of the River Thames, there was an extensive cemetery in Southwark with dispersed burials

(Dean and Hammerson, 1980; Hall, 1996). For additional information, see the Centre for Human Bioarchaeology at the Museum of London (MOL) and from the Prescot Dig Online Compendium (n.d.).

A total of 10 complete femora from each of three cemeteries were obtained from the MOL with the collaboration and assistance of then-director Bill White (now deceased). Skeletons from three Roman burial sites, designated Eastern, Western and Southern, were selected from the archives by White in association with two of the co-authors (M.M.S. and P.M.F.). With one exception (MOL# 4STS82-886), a burial from a pit that White considered to be an early outlier, all the skeletons were expected to represent the second and third centuries of Roman habitation during the peak of Londinium's urban development. Most skeletons had been sexed and other than visual examination, the bones had no prior processing or treatment; all were stored in the museum archive in cardboard boxes. Many of the bones had the original burial soil still adhered to them, and in some cases the soil was loose in the box.

Pre-Roman, Iron Age Danebury

A total of 28 bone samples from the Danebury Iron Age hillfort in Hampshire (south central England) were provided by Barry Cunliffe assisted by David Allen. Danebury, covering 5 ha, is considered the typical developed hillfort and was excavated by Cunliffe between 1969 and 1988 (Cunliffe, 1984, 1995; Cunliffe and Poole, 1991). Established in the sixth century BC, the hillfort was in active use for over 500 years. During that time, at least 300 individual deposits of human remains were made, primarily in abandoned grain storage pits. Selection of specimens was essentially random as described by Cunliffe *et al.* (2015).

Pre-Roman, Iron Age East Yorkshire

Bone samples from three pre-Roman Iron Age cemeteries in East Yorkshire were provided by the British Museum through the good offices of J. D. Hill. Five samples from Kirkburn, 14 samples from Burton-Fleming and 19 samples from the Rudston cemetery, giving a total of 38 specimens, were analysed. Hundreds of burials were unearthed during the period 1967–87 at these sites (Stead, 1991), belonging to a distinctive local group known as the Arras culture (Jay *et al.*, 2012; Peck *et al.*, 2013), characterized by large cemeteries of inhumations often within rectangular ditched enclosures, a method unique in Britain, but with similarities to burial rites practised on Continental Europe. Samples for this study were chosen to include the best quality femora from each site.

Femur and burial soil processing

Methods used to obtain bone cores, clean them and extract minerals, as well as perform inductively coupled plasma mass spectrometry (ICPMS) in a High efficiency particulate air (HEPA)-filtered, dedicated trace metal clean laboratory (Hurley *et al.* 1996; Shafer *et al.* 1999), were exactly as described by Shafer *et al.* (2008). Burial soil adhering to the femora was recovered with the aid of clean Teflon implements and was processed and analysed as described by Shafer *et al.*

Femur elemental analysis

Pb and a suite of 24 additional trace (Ag, Al, As, Cd, Ce, Co, Cr, Cu, Fe, La, Li, Mn, Ni, Pt, Rb, Ti, V, U, Zn) and bone-forming elements (Ba, Ca, Mg, P, Sr) were quantified in the bone core

digests using a Finnigan Element 2 magnetic sector ICPMS (Shafer *et al.* 2008). Three internal standards (Ga, In and Ir) were spiked into the diluted digests before analysis. Each batch of 22 cores included three NIST 1400 (National Institute of Standards and Technology, Gaithersburg, MD, USA) bone ash-certified reference material samples, three multi-element sample digestion spike recovery samples, several sample duplicates and four method blanks. Each ICPMS run included matrix spike recoveries, analytical replicates, instrument blanks, and isotope and resolution checks. Multiple isotopes (where feasible) of many of the target elements were acquired as part of the overall data quality assessment. These data were used to examine bone integrity and correct for soil contamination after Shafer *et al.* (2008).

Femora and soil: Pb and Sr isotopic compositions

To measure Pb and Sr isotopic compositions, we selected skeletons from each of three Londinium cemeteries to include representation of both sexes, the entire range of Pb concentrations and variation in the estimated age of death of the individuals. Additionally, a subgroup of samples was similarly selected from the Danebury (10 samples) and East Yorkshire (21 samples) populations. The digests of these subgroups were further processed to produce purified fractions of Pb and Sr. Bone standard reference materials NIST 1400 (bone ash) and NIST 1486 (bone meal), USGS rock standards BCR-2 (basalt) and AGV-2 (andesite), and soil standard NIST 2709 (San Joaquin soil) were included in these purification protocols for quality assurance.

Sample preparation

Bone and soil digests (described previously) were subsampled such that the aliquot contained > 50 ng Pb and evaporated to dryness in a Teflon beaker (Savillex Corp., Eden Prairie, MN, USA). The samples were reconstituted in 2 M HNO₃ and the Sr and Pb fractions were separated by passing through Sr Spec resin (10–20 µm, Eichrom Technologies, Inc., Lisle, IL, USA). The resin was pre-cleaned in 0.2 M HNO₃, then stored and loaded into the column as a H₂O slurry. Analyte fractions were separated from the sample matrix using a method modified from Deniel and Pin (2001), Irrgeher *et al.* (2013) and Liu *et al.* (2016) (see Table S1 in the additional supporting information for column separation procedure). Sr and Pb fractions were reconstituted in 16 M, Optima-grade HNO₃ before analysis.

Pb and Sr isotope analysis

All isotopic analyses were performed on a Neptune Plus MC-ICPMS (Thermo Fisher Scientific Inc., Waltham, MA, USA) housed in the Trace Elements Clean Lab at the Wisconsin State Laboratory of Hygiene. Pb isotopic composition measurements followed the methods of Smith *et al.* (2017), Kuritani and Nakamura (2002), and Tanimizu and Ishikawa (2006). Tl was added to each sample to correct for mass bias using the exponential fractionation law and assuming $^{205}\text{Tl}/^{203}\text{Tl} = 2.387$. Pb and Tl isotopes were analysed in static mode using six Faraday collectors; a seventh collector was used to monitor and correct for Hg interferences. All samples, NBS981 bracketing standards and reference standards were blank-corrected (2% v/v HNO₃). Pb and Tl concentrations (in 2% HNO₃) in the final analyte solutions were targeted at either 10 ng g⁻¹ (ppb) Pb + 2 ng g⁻¹ Tl or 40 ng g⁻¹ Pb + 10 ng g⁻¹ Tl. Isotope ratios are normalized to the days average of NBS981 ratio for each individual analytical day with values reported relative to those NBS981 ratios determined by Thirlwall (2002).

Sr isotopes were analysed following the method of Konter and Storm (2014) using four Faraday collectors, with additional collectors used to monitor and correct for Kr and Rb isobaric interferences. ^{85}Rb was monitored to correct for any ^{87}Rb interfering at mass 87 and ^{83}Kr was monitored to correct for any contributions from ^{84}Kr and ^{86}Kr present in the Ar source. Mass bias corrections were made by monitoring the internal $^{88}\text{Sr}/^{86}\text{Sr}$ ratio using the exponential law and assuming the $^{86}\text{Sr}/^{88}\text{Sr}$ value of 0.1194 determined for NBS987 by Wise and Watters (2007). Only the Apex HF desolvating nebulizer was used for introduction of the Sr sample fractions. Samples, NBS987 bracketing standards and reference standards were prepared to a final concentration of 50 ng g^{-1} total Sr in 2% v/v HNO_3 . Isotope ratios are normalized to the days average of NBS987 for each individual analytical day with values reported relative to $^{87}\text{Sr}/^{86}\text{Sr} = 0.71024$ (Kagami *et al.*, 1982).

RESULTS

Analytical performance

General trace element analytical method performance was as detailed in Shafer *et al.* (2008). Pb and Sr stable isotope results for NIST1400 (bone ash), NIST1486 (bone meal), NIST2709a (San Joaquin soil), BCR-2 (Columbia River basalt), and AGV-2 (Guano Valley andesite) are tabulated in additional supporting information Table S2 and agree well with published values (additional supporting information Fig. S1). External reproducibility at 2 SD for Sr isotopic compositions of these standards is typically $< 20\text{ ppm}$, with NIST 1486 being the exception at 46 ppm. External reproducibility at 2 SD for Pb isotopic compositions of the same standards is on the order of 0.1–0.15% for the $^{206}\text{Pb}/^{204}\text{Pb}$, $^{207}\text{Pb}/^{204}\text{Pb}$, and $^{208}\text{Pb}/^{204}\text{Pb}$ ratios, and 0.05–0.1% for the $^{207}\text{Pb}/^{206}\text{Pb}$ and $^{208}\text{Pb}/^{206}\text{Pb}$ ratios.

^{14}C ages of bone specimens

Radiocarbon dates are listed in Table 1. For the East Yorkshire sites, all but one specimen dated to between 2305 ± 20 and $2145 \pm 20\text{ RC yr. BP}$ (405–105 BC). A younger age outlier was Specimen R175, which dated to $2035 \pm 20\text{ RC yr. BP}$ (110 cal BC–25 cal AD) (UCIAMS-11336). These results confirm and extend the data reported by Jay *et al.* (2012).

At the Danebury site, five of six specimens dated between 2361 and 2184 RC yr. BP (Cunliffe *et al.* 2015) (Table 1). The sixth specimen, D3, was older and dated to $2389 \pm 26\text{ RC yr. BP}$ (OxA-25953). Nine of 10 Londinium burials dated from 1876 ± 25 to $1713 \pm 25\text{ RC yr. BP}$ (70–395 cal AD), an interval consistent with the Roman occupation. The pit burial outlier from Londinium, femur #4STS82-886, dated to $2009 \pm 26\text{ RC yr. BP}$ (90 cal BC–60 cal AD) (OxA-20197).

Geochemical and isotopic compositions of bones and soils

Major and trace element composition of the Roman-age bones can be found in Table S3 in the additional supporting information. The soil-corrected concentrations of Pb in Roman (Londinium-era) bone specimens ranged from 8 to $123\text{ }\mu\text{g g}^{-1}$ (median = $26.5\text{ }\mu\text{g g}^{-1}$; $n = 27$) (Fig. 1), greater than that measured in Iron Age bones from Danebury (median = $0.31\text{ }\mu\text{g g}^{-1}$; range = 0.082–1.7; $n = 28$) and East Yorkshire (median = $0.42\text{ }\mu\text{g g}^{-1}$; range = 0.07–2.9; $n = 38$). With the few outliers excluded, pre-Roman Iron Age bone Pb concentrations do not exceed $0.5\text{ }\mu\text{g g}^{-1}$. The femur from the older, pre-Roman pit burial showed a much lower Pb concentration of $0.21\text{ }\mu\text{g g}^{-1}$.

Table 1 ^{14}C ages of femur samples measured by accelerator mass spectrometry (AMS)

Cemetery	Sample number	$^{14}\text{C} \pm \text{SD RC}$ yr BP	$\delta^{13}\text{C}_{\text{‰}}$ (VPDB)	$\delta^{15}\text{N}_{\text{‰}}$ (AIR)	cal AD/BC 2 SD (95.4% CI)	AMS laboratory number
<i>Londinium</i>						
Eastern	H0088-829	1808 $\pm$ 25	−18.9		130–320 cal AD	OxA-19476
Eastern	H0088-942	1876 $\pm$ 25	−18.5	9.7	70–220 cal AD	OxA-20194
Southern	CB83-2354	1801 $\pm$ 24	−19.3		130–325 cal AD	OxA-19477
Southern	GVD96-70	1864 $\pm$ 26	−18.5	10.6	75–225 cal AD	OxA-20198
Southern	4STS82-886	2009 $\pm$ 26	−19.8	12.5	90 cal BC–60 cal AD	OxA-20197
Western	WES89-421	1650 $\pm$ 25	−19.6	8.6	335–530 cal AD	OxA-20195
Western	WES89-421	1713 $\pm$ 25	−19.1		250–395 cal AD	OxA-20363
Western	WES89-487	1723 $\pm$ 24	−19.3		250–390 cal AD	OxA-19474
Western	WES89-487	1735 $\pm$ 23	−19.3		240–385 cal AD	OxA-19475
Western	WES89-686	1832 $\pm$ 26	−18.7	10.6	85–245 cal AD	OxA-20196
<i>East Yorkshire</i>						
Burton Fleming	BF6	2240 $\pm$ 20	−20.5	10.8	385–205 cal BC	UCIAMS-11325
Burton Fleming	BF20	2305 $\pm$ 20	−20.4	10.6	405–360 cal BC	UCIAMS-11326
Garton Station	GS6	2145 $\pm$ 20	−20.3	10.9	355–105 cal BC	UCIAMS-11327
Kirkburn	K3	2175 $\pm$ 20	−20.3	10.9	360–170 cal BC	UCIAMS-11328
Kirkburn	K5	2210 $\pm$ 20	−20.3	9.3	365–200 cal BC	UCIAMS-11330
Rudston	R22	2250 $\pm$ 30	−20.4	10.6	395–205 cal BC	UCIAMS-11331
Rudston	R106	2250 $\pm$ 25	−20.5	10.2	395–205 cal BC	UCIAMS-11332
Rudston	R140	2270 $\pm$ 25	−20.2	10.5	400–210 cal BC	UCIAMS-11333
Rudston	R143	2280 $\pm$ 25	−20.3	10.4	405–230 cal BC	UCIAMS-11335
Rudston	R175	2035 $\pm$ 20	−20.2	9.4	110 cal BC–25 cal AD	UCIAMS-11336
<i>Danebury*</i>						
	D3	2389 $\pm$ 26	−20.0		703–397 cal BC	OxA-25953
	203	2147 $\pm$ 26	−20.0		355–92 cal BC	OxA-25954
	209	2157 $\pm$ 26	−20.0		356–107 cal BC	OxA-25955
	215	2162 $\pm$ 27	−19.9		358–112 cal BC	OxA-25956
	234	2192 $\pm$ 27	−20.2		361–184 cal BC	OxA-25957
	240	2133 $\pm$ 26	−20.0		350–56 cal BC	OxA-25958

AMS radiocarbon dates on human femora are from Londinium cemeteries. ^{14}C ages are given in radiocarbon years BP; calibrated ages are calculated using OxCal 4.3 (IntCal 13), C. Bronk Ramsey 2019, Interface build 118, Updated 25 April 2019.

*Danebury ages from Cunliffe *et al.* (2015); published calibrated ages revised to OxCal 4.3 (IntCal 13).

Pb and Sr isotopic compositions of femora are presented in Table 2. Pb isotopic compositions of Roman femora show a relatively small range, and are comparable with local British and German Pb ore isotopic compositions, and previously published enamel results from Roman and medieval enamel (Fig. 2). In contrast, pre-Roman bone samples contain a larger range in Pb isotopic

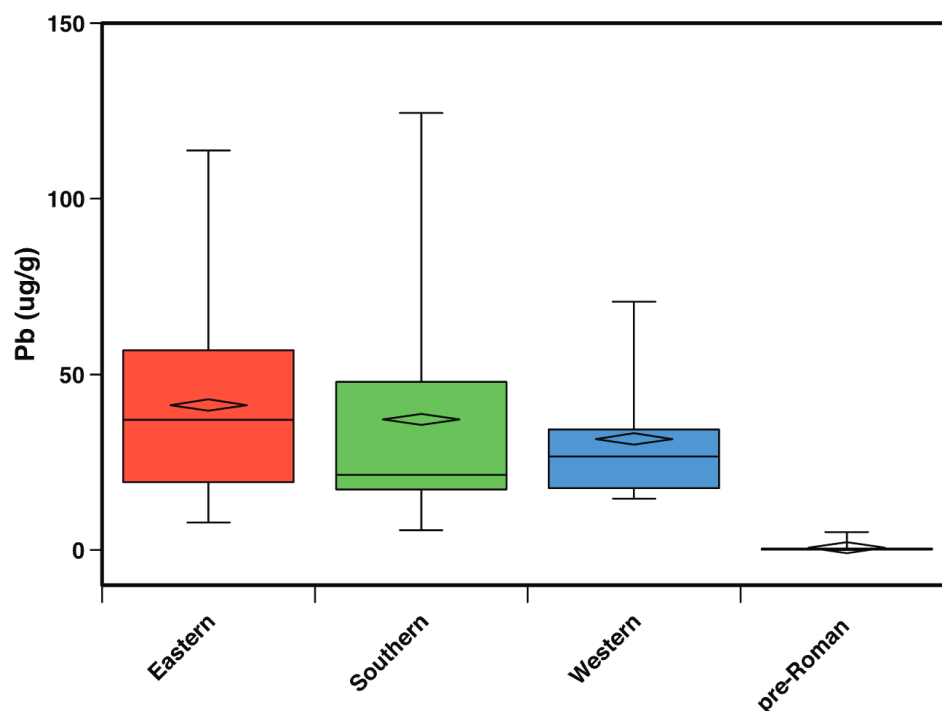


Figure 1 Box and whisker plots illustrating the elevated lead (Pb) levels of Roman specimens from all three cemeteries compared with pre-Roman populations. The means of each data set are also shown. These data show that Pb exposure increased significantly between the pre-Roman and Roman occupational periods in Londinium. [Colour figure can be viewed at wileyonlinelibrary.com]

composition. Pb isotopic compositions of some burial soils were also analysed and show similar compositions to the bones and enamel from the same time period.

DISCUSSION

Provenance of skeletons

AMS radiocarbon ages agree well with the predicted ages based on the archaeological evidence. The older ages of the East Yorkshire and Danebury specimens confirm that they represent a pre-Roman Iron Age population. The present measured Sr isotope values of the Londinium bone samples range from $^{87}\text{Sr}/^{86}\text{Sr} = 0.7087$ to 0.7099 , consistent with the Londinium geological and hydrological signature (Trickett *et al.*, 2003; Evans *et al.*, 2010, 2012; Shaw *et al.*, 2016). However, the input of the measured range of $^{87}\text{Sr}/^{86}\text{Sr}$ into the Biosphere Isotope Domains search module (2019) indicates these Sr isotope values cannot definitively place the specimens in Londinium. Furthermore, diagenetic alteration of the Sr content and isotopic composition cannot be ruled in the data set. Nevertheless, the ^{14}C ages of the specimens demonstrate that these people lived during the period associated with the Roman occupation of London and would have been subject to the environmental conditions of the time.

Procedural validation and bone integrity

To use trace element geochemistry of archaeological bone as a record of exposure, the potential for diagenetic alteration must be addressed for the elements of interest. We evaluate the fidelity of

Table 2 Strontium (Sr) and lead (Pb) isotopic compositions of Roman and Celtic bones measured by multi-collector inductively coupled plasma mass spectrometer (MC-ICP-MS)

Site number	Skeleton	$^{87}\text{Sr}/^{86}\text{Sr}$	Error	$^{208}\text{Pb}/^{204}\text{Pb}$	$^{207}\text{Pb}/^{204}\text{Pb}$	$^{206}\text{Pb}/^{204}\text{Pb}$	$^{207}\text{Pb}/^{206}\text{Pb}$	$^{208}\text{Pb}/^{206}\text{Pb}$
<i>Roman, Eastern group</i>								
HOO88	416	0.709344	0.000011	38.642	15.689	18.517	0.84727	2.08682
	<i>dup.</i>	0.709353	0.000010	38.613	15.678	18.505	0.84725	2.08668
HOO88	459	0.709836	0.000008					
	<i>dup.</i>	0.709843	0.000013					
HOO88	605			38.610	15.681	18.496	0.84781	2.08746
HOO88	829	0.709089	0.000013	38.588	15.682	18.475	0.84881	2.08870
HOO88	835							
HOO88	942	0.709547	0.000015	38.565	15.670	18.443	0.84964	2.09101
	<i>dup.</i>			38.569	15.672	18.445	0.84969	2.09107
HOO88	1022							
HOO88	1213	0.709340	0.000010	38.541	15.672	18.487	0.84769	2.08476
HOO88	1410	0.709573	0.000007	38.621	15.682	18.501	0.84763	2.08748
HOO88	1482	0.709679	0.000005	38.615	15.681	18.508	0.84724	2.08644
	<i>dup.</i>	0.709680	0.000008	38.617	15.682	18.510	0.84722	2.08639
<i>Roman, Southern group</i>								
4STS82	886	0.708879	0.000007	38.588	15.644	18.586	0.84166	2.07608
CB81	2206	0.708993	0.000016	38.487	15.657	18.455	0.84840	2.08546
	<i>dup.</i>			38.474	15.651	18.450	0.84833	2.08541
CB83	2354	0.709214	0.000010	38.474	15.652	18.449	0.84838	2.08547
COSE84	1355	0.709192	0.000013	38.477	15.652	18.463	0.84774	2.08403
COSE84	1834	0.709366	0.000009	38.415	15.633	18.425	0.84850	2.08495
	<i>dup.</i>	0.709376	0.000010					
	<i>dup.</i>	0.709373	0.000010					
	<i>dup.</i>	0.709353	0.000008					
	<i>dup.</i>	0.709362	0.000007					
COSE84	1951	0.708680	0.000010					
COSE84	96			38.712	15.676	18.558	0.84470	2.08600
	<i>dup.</i>			38.721	15.691	18.563	0.84469	2.08595
GDV96	70	0.708859	0.000012					
	<i>dup.</i>	0.708871	0.000019					

(Continues)

GDV96	126	0.709334	0.000018						
	dup.	0.709322	0.000014						
GDV96	242	0.708671	0.000010	38.669	15.658	18.577	0.84286	2.08150	
	dup.			38.660	15.656	18.576	0.84284	2.08126	
GDV96	360	0.709106	0.000008						
Roman, Western group									
WES89	398	0.709316	0.000008	38.422	15.632	18.456	0.84697	2.08186	
	dup.			38.427	15.633	18.458	0.84698	2.08191	
WES89	421	0.709402	0.000008	38.479	15.651	18.467	0.84752	2.08370	
	dup.			38.482	15.652	18.469	0.84754	2.08372	
WES89	482	0.709339	0.000008						
WES89	485	0.709228	0.000009						
	dup.	0.709233	0.000007						
WES89	487	0.709779	0.000009	38.501	15.656	18.478	0.84729	2.08362	
WES89	504	0.709375	0.000010	38.532	15.664	18.465	0.84831	2.08675	
	dup.			38.529	15.663	18.464	0.84832	2.08675	
WES89	599	0.709262	0.000009	38.442	15.633	18.440	0.84778	2.08480	
	dup.			38.444	15.633	18.439	0.84780	2.08490	
WES89	686	0.709209	0.000009	38.678	15.676	18.533	0.84585	2.08701	
	dup.			38.683	15.678	18.536	0.84583	2.08697	
WES89	837	0.709452	0.000010						
	dup.	0.709442	0.000010						
Celtic, Danebury									
284b	DA72 p120 A			38.742	15.681	18.851	0.83188	2.05528	
297	DA78 p1078 A			38.552	15.665	18.587	0.84279	2.07416	
	dup.			38.556	15.667	18.589	0.84278	2.07414	
302	DA76 p923 A			38.609	15.667	18.638	0.84061	2.07155	
1262	DA83 p2218 A			38.780	15.676	18.814	0.83324	2.06125	
1262	DA72 p343 A			38.722	15.681	18.814	0.83344	2.05809	
161D	DA80 p1545 A			38.804	15.688	18.925	0.82893	2.05037	
161D	DA74 p582 A			38.738	15.677	18.844	0.83195	2.05576	
296	DA72 p382 A			38.790	15.682	18.946	0.82774	2.04748	
308	DA76 p829 A			38.822	15.689	18.955	0.82769	2.04820	
1266	DA85 p2462			38.760	15.688	18.840	0.83270	2.05731	

(Continues)

Celtic, East Yorkshire

R30	38.868	15.690	18.769	0.83600	2.07095
R102	38.587	15.654	18.591	0.84206	2.07566
BF63	38.993	15.695	18.764	0.83643	2.07810
<i>dup.</i>	38.998	15.696	18.766	0.83641	2.07813
R140	38.713	15.655	18.750	0.83499	2.06480
R81	38.860	15.656	18.747	0.83517	2.07294
K6	38.812	15.803	18.429	0.85753	2.10612
<i>dup.</i>	38.809	15.802	18.428	0.85753	2.10609
K82	38.725	15.662	18.688	0.83807	2.07226
BF24	38.647	15.660	18.635	0.84038	2.07399
K4	38.653	15.667	18.622	0.84134	2.07575
BF46	38.714	15.665	18.753	0.83531	2.06446
BF25	38.828	15.820	18.401	0.85978	2.11018
R162	38.829	15.809	18.433	0.85762	2.10652
K5 1–2	38.538	15.662	18.513	0.84601	2.08176
BF1	38.592	15.643	18.627	0.83981	2.07193
R107	38.810	15.677	18.651	0.84054	2.08093
<i>dup.</i>	38.819	15.679	18.655	0.84049	2.08094
R78	38.822	15.802	18.434	0.85720	2.10604
R97	38.794	15.662	18.743	0.83564	2.06982
R104	38.906	15.691	18.749	0.83689	2.07518
<i>dup.</i>	38.906	15.690	18.748	0.83691	2.07525
BF13	38.646	15.709	18.548	0.84693	2.08361
BF42	38.677	15.665	18.772	0.83447	2.06041
R144	38.854	15.685	18.634	0.84172	2.08519
<i>dup.</i>	38.864	15.688	18.638	0.84173	2.08530

See Table S2b in the additional supporting information for estimates of Pb isotope uncertainty.

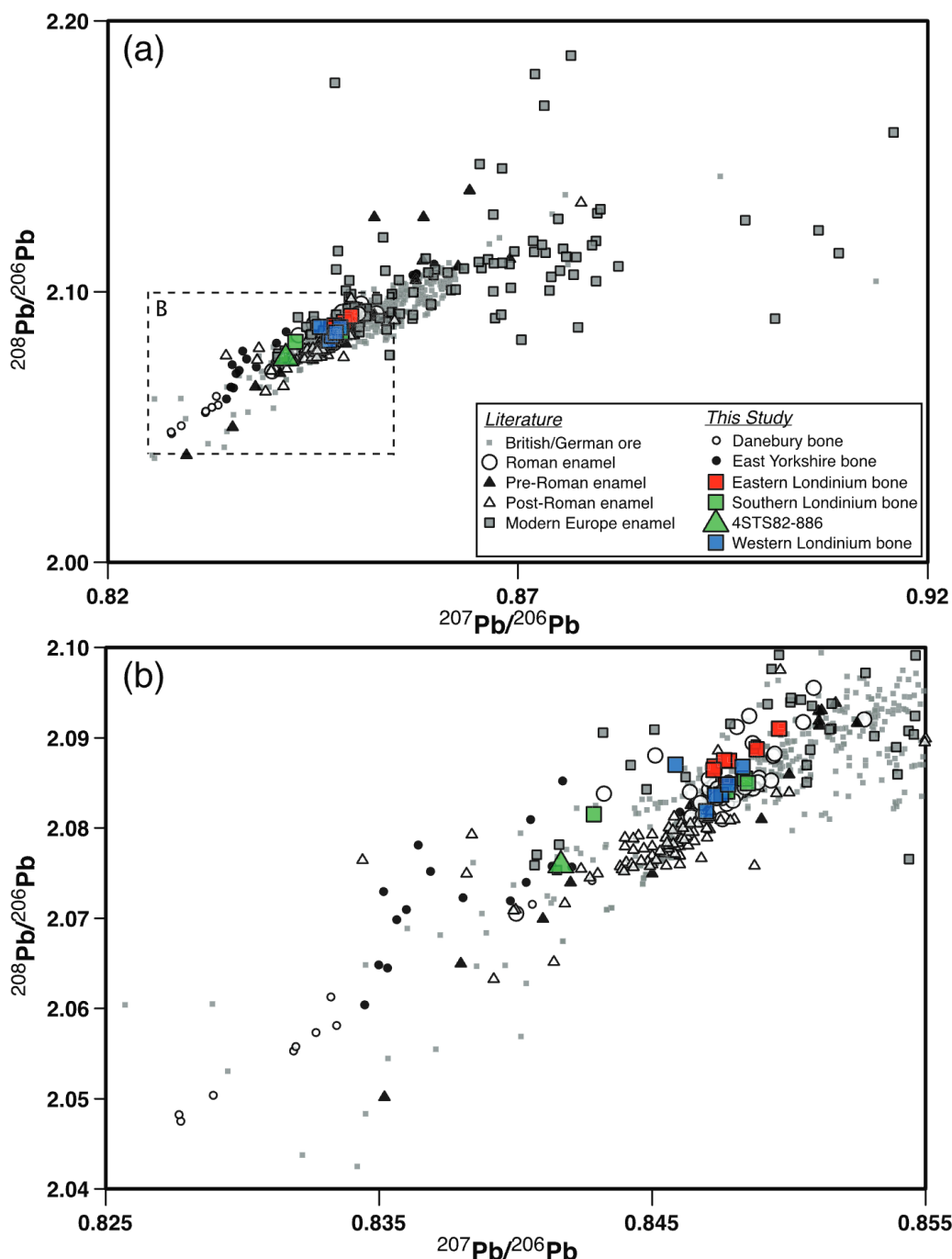


Figure 2 Lead (Pb) isotopic compositions of Roman and pre-Roman bones compared with German and British ore data (Haggerty, Budd, Rohl, & Gale, 1996; Rohl 1996; Durali-Mueller, Brey, Wigg-Wolf, & Lahaye 2007) and European enamel data compiled by Kamenov and Gulson (2014) and more recent data (Shaw et al., 2016): (a) $^{208}\text{Pb}/^{206}\text{Pb}$ versus $^{207}\text{Pb}/^{206}\text{Pb}$, with a zoomed in plot in (b). The majority of Pb isotopic compositions for Roman specimens fall within the range of ore compositions, and cluster around a narrow range in isotope composition of $^{207}\text{Pb}/^{206}\text{Pb} \sim 0.845\text{--}0.849$, the signal of widespread lead pollution proposed by Montgomery, Evans, Chenery, Pashley, & Killgrove, (2010). In contrast, pre-Roman samples span a significantly wider range and likely represent highly localized, minor sources of Pb. Some enamel samples deviate from the typical Londinium ore signal isotopic composition, and these deviations were used by Shaw et al. (2016) to identify potential migrants to Londinium. However, these isotopic ratios are not significantly different from the pre-Roman population compositions, suggesting that these migrants may have come from relatively proximal regions of the Roman Empire (i.e., Britain). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the geochemical composition of the bone with multi-element characterization, specifically focusing on the Pb contents and isotopic compositions of the bone. Although recent experimental work has shown that Pb is not significantly absorbed into bone in a variety of burial environments (Krajcarz, 2019), the potential for Pb uptake during burial must still be evaluated.

First, the Ca and P contents and Ca/P ratios of the femora are well within the expected values compared with hydroxyapatite chemistry of modern unaltered bones (Fig. 3, a) (Tzaphlidou and Zaichick, 2003). In addition, Zn concentrations of the majority of femora fall within the range expected for modern human bone at 140–560 $\mu\text{g g}^{-1}$, (Brown and Blakely, 1985). However, these metrics do not necessarily preclude the influence of diagenetic alteration on other elements, as different elements behave differently in the burial environment (Krajcarz, 2019). Therefore, the properties of the soil must first be considered to evaluate the potential for Pb mobility in the burial environment.

Pb is strongly partitioned and retained by soils (Sauve, Hendershot, & Allen, *et al.*, 2000; Shepard *et al.*, 2009) and previous work on Pb contents in archaeological bone suggested that, if anything, Pb would be lost from the bone rather than absorbed (e.g., Waldron *et al.*, 1976). However, Pb mobility in soils is strongly related to pH (Sauve *et al.*, 2000), with Pb being more mobilized as pH decreases. The measured pHs of the burial soils from the sites in this study range from 6.81 to 8.66, and calculated partition coefficients based on the average pH-partitioning relationship in Sauve *et al.* (2000) indicate that Pb is highly immobile in these burial soils ($\log K_d = 4.7\text{--}5.6$; additional supporting information Table S4). Thus, the properties of the Londinium soils suggest that Pb was unlikely to be mobilized in the burial environment. Regardless of the soil properties, diagenetic alteration of the bones is unavoidable (Krajcarz, 2019), and elements with a known affinity for bone uptake can be used to determine whether or not Pb shows similar behaviour (i.e., if Pb has been elevated by diagenetic alteration).

It is well established that the uptake of environmental U occurs during burial (e.g., Millard and Hedges, 1995, 1996), and U contents of the femora (0.3–13.1 $\mu\text{g g}^{-1}$) are elevated compared with expected value based on unaltered tooth enamel (about 0.05 $\mu\text{g g}^{-1}$; Kamenov *et al.*, 2018). In addition, recent experimental work documenting changes in bone chemistry during burial indicate that Mn is rapidly absorbed into bone regardless of the burial environment (Krajcarz, 2019). Thus, U and Mn can be used as proxies for chemical alteration due to diagenesis; if Pb is correlated with either of these elements, then it is logical to conclude that the Pb contents were modified by diagenesis. However, if the Pb contents of the bones are not correlated to the U and/or Mn contents, then it is logical to conclude that the Pb contents were not acquired through diagenesis, but rather were acquired by *in vivo* exposure to elevated levels of Pb. Furthermore, previous work using other elements to evaluate diagenesis (e.g., Ba; Patterson *et al.*, 1987) can also provide an insight into potential diagenetic processes affecting the Roman femora.

In Figure 3 (b), Ba/Ca values as a function of U/Ca in the Roman femora show a clear signal for diagenetic alteration with respect to these elements of special interest. However, in Figure 3 (c, d), Pb/Ca as a function of U/Ca and Mn/Ca show no such correlation. If Pb were subject to post-burial modification as are U and Mn, a strong correlation would exist between the U and Pb and Mn and Pb contents of the bone. The lack of correlation in Figure 3 (c, d) strongly demonstrates that Pb content has not been significantly changed by post-burial processes. In contrast, Ba/Ca is positively correlated with U/Ca (Fig. 3, b), suggesting that Ba content is at least partly a diagenetic product, consistent with the inference shown by Patterson *et al.* (1987). In summary, our multi-element data provide proof that Pb measured in the human bones is biologically, not secondarily, derived.

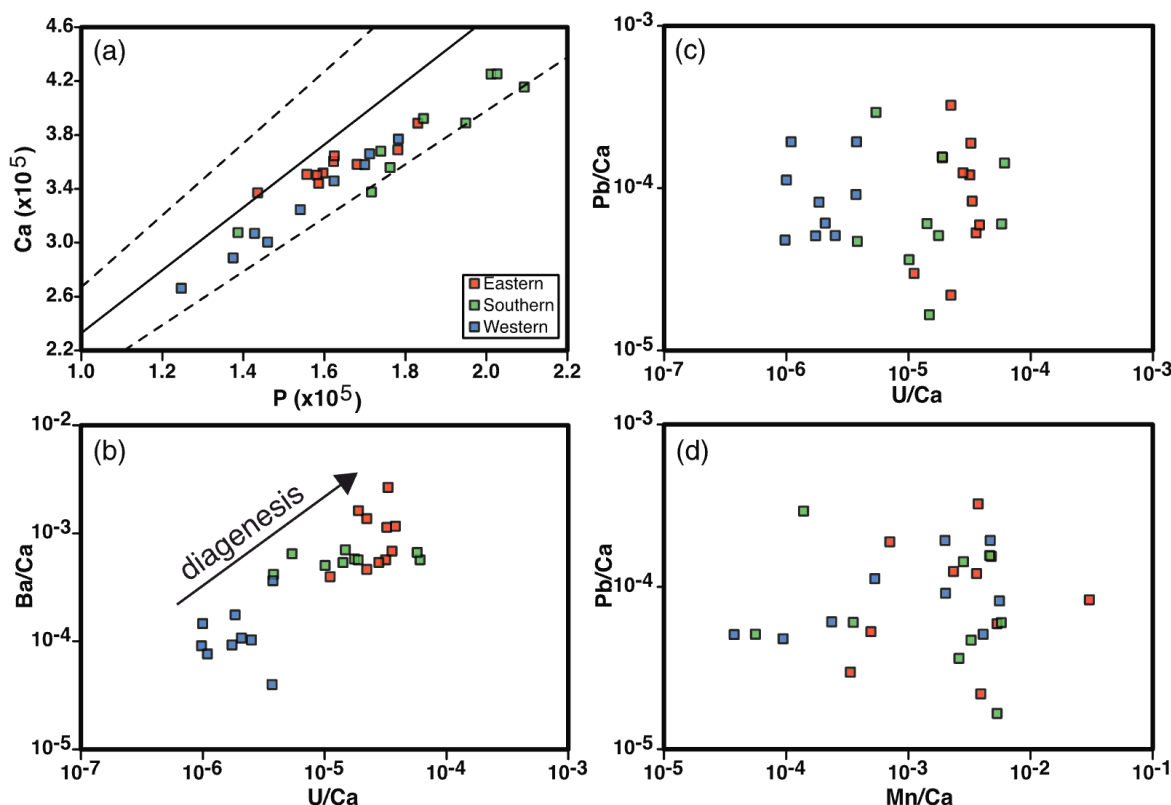


Figure 3 Major and trace element systematics of Roman bones: (a) calcium (Ca) and phosphorus (P) concentrations of Roman bones measured in this study. The solid and dashed lines represent the Ca/P ratio and uncertainty (2.33 ± 0.34) for modern human rib bones measured by Tzaphlidou and Zaichick (2003). (b) Ba/Ca versus U/Ca, showing that both barium (Ba) and uranium (U) increase simultaneously, suggesting that both elements are subject to alteration during diagenesis. Diagrams are modelled after the diagenetic analysis approach of Patterson et al. (1987). (c) Pb/Ca versus U/Ca, showing that increases in the U content of bone due to post mortem uptake are not correlated to Pb content, indicating that the addition of Pb during diagenesis is minimal in the Roman femora. (d) Similar to (c), a lack of correlation between Pb/Ca and Mn/Ca (manganese) indicates that the addition of Pb during diagenesis is minimal in the Roman femora. [Colour figure can be viewed at wileyonlinelibrary.com]

Pb concentrations in Roman and pre-Roman bone specimens

Although many Roman cities in Britain developed on the site of Iron Age *oppida*, the port city of Londinium appears to have been a *de novo* urban development (Wilkes, 1996). There is, however, ample evidence of pre-Roman occupation along the lower Thames Valley. The pit burial (sample 4STS82-886) from near the Southern Cemetery is an example (Hall, 1996). It provides strong evidence that the Iron Age people living in the same environment in which Londinium was to develop did not have high Pb concentrations in their bone. These individuals' Pb levels are comparable with those of the pre-Roman Iron Age people sampled from Danebury and the Eastern Yorkshire cemeteries, all of whom showed no evidence of exposure to Pb and other toxic metals. In addition to those discordances, further evidence of the extraordinarily high concentrations of Pb in the Londinium cohort can be seen when compared with our data from other archaeological sites such as Austrian Iron Age cemeteries (Shafer et al. 2008), where we found a median Pb concentration in 80 femora of $0.23 \mu\text{g g}^{-1}$ (range = $0.01\text{--}5.5 \mu\text{g g}^{-1}$), which is similar to that observed in the pre-Roman Iron Age specimens of Britain.

Source(s) of Pb in Roman and Iron Age bones

The high-precision bone Pb isotope data (Fig. 2) reveal that the $^{207}\text{Pb}/^{206}\text{Pb}$ and $^{208}\text{Pb}/^{206}\text{Pb}$ ratios in Roman and pre-Roman populations of Britain reflect contrasting origins of Pb. Pb isotopic compositions in Roman femora exhibit a narrow range, with average $^{207}\text{Pb}/^{206}\text{Pb} = 0.8472 \pm 0.0033$ and $^{208}\text{Pb}/^{206}\text{Pb} = 2.0856 \pm 0.0048$ (2σ , $n=30$). The average $^{207}\text{Pb}/^{206}\text{Pb}$ measured in our specimens is within the range of Roman individuals designated to have ‘polluted’ Pb in their tooth enamel ($^{207}\text{Pb}/^{206}\text{Pb} = 0.845\text{--}0.849$; Montgomery *et al.*, 2010). Furthermore, the Pb isotope ratios of ‘polluted’ Romans (Montgomery *et al.*, 2010) and the present bone results are consistent with the widespread atmospheric deposition signal during the Roman period (Fig. 4). As such, our specimens were likely exposed to the same widespread anthropogenic Pb pollution as those previously studied.

The Pb isotope data show that Pb exposures of the Roman population of Londinium occurred during the first to fourth centuries AD and likely came from the same sources as other Roman specimens (e.g., Montgomery *et al.*, 2010). Although *post mortem* exposures have been suspected previously (Scarborough, 1984; Patterson *et al.*, 1987), this is quite unlikely based on our chemical assessment protocol, the lack of contaminating grave goods (Thompson, Westman, & Dyson, 1998) and the absence of Pb coffins (Schofield & Maloney, 1998). Several specific exposure vectors are possible, including occupational exposure (e.g., Craddock 1979), consumption of adulterated food, wine and/or contaminated water (e.g., Gilfillan, 1965; Delile

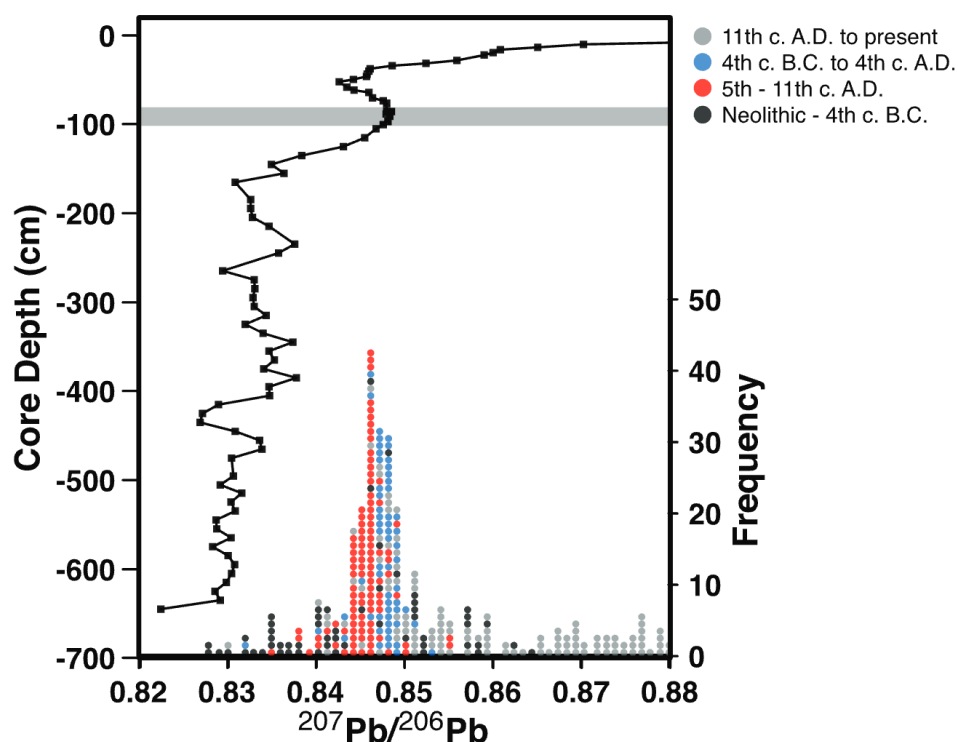


Figure 4 Lead (Pb) isotope distribution of enamel (literature) and bones (present study) compared with a Belgian peat bog core (Allan *et al.*, 2018). Note that we use only a single peat bog core for simplicity; however, there are many peat bog core records that show a similar Pb isotopic composition during the Roman period (e.g., Shotyk *et al.*, 1998). The major isotopic peak in humans corresponds to a similar ratio in peat bog core records and indicates that a widespread source of Pb contributed to Pb exposure during the Roman period and likely continued into medieval times. [Colour figure can be viewed at wileyonlinelibrary.com]

et al., 2014), 'urban island' effects similar to those found in modern times (Lee *et al.*, 2006), or general atmospheric pollution associated with elevated atmospheric Pb levels during the Roman period (McConnell *et al.*, 2018). Another likely source is the Romans consumption of Pb-contaminated salt produced from evaporation or boiling of the readily available British brine in Pb pots and pans (Penney and Shotter, 1996, 2001). When parsed by sex (additional supporting information Fig. S2), both Pb concentrations and Pb isotope ratios are not significantly different between males and females, implying that the source/usage practices of Pb were not gender specific during that era. As such, our data provide further evidence for a widespread pollution signal that cannot be attributed to any single practice.

Implications of elevated Pb concentrations in Roman bones

One of the most intriguing and unsettled questions in Roman archaeological studies has been whether or not the Romans were subjected to poisoning from Pb. While some have even argued that the collapse of the Roman Empire was due to widespread Pb poisoning (e.g., Gilfillan, 1965; Nriagu, 1983), others have rejected this hypothesis (e.g. Hodge 1991; Scarborough, 1984; Retief & Cilliers, 2005). Measurements of Pb in tooth enamel, while providing evidence for Pb exposure, do not provide any quantitative metrics for the true vector of exposure: blood Pb level. Our data showing high concentrations of Pb in femora can be used to address the potential for elevated blood Pb levels in Romans of Londinium. The US National Institute for Occupational Safety and Health (NIOSH) has designated the reference blood Pb level for case definition of toxicity at $5 \mu\text{g dl}^{-1}$ in adults. The average blood Pb level among adults residing in the United States, as reported in the national report on human exposure to environmental chemicals (National Health and Nutrition Examination Survey, Center for Disease Control, see web references), was $0.97 \mu\text{g dl}^{-1}$ from 2013–2014 (Agency for Toxic Substances and Disease Registry, Centers for Disease Control and Prevention (CDC), 2018). Blood Pb levels are considered 'very high' when they exceed $40 \mu\text{g dl}^{-1}$.

Several studies have attempted to measure and relate blood Pb levels to Pb concentrations in bone. In a pioneering study on occupational Pb exposure in Pb smelter employees, Flemming *et al.* (1997) established the cumulative blood Pb index, which helps to predict bone Pb concentration based on long-term exposure to Pb. Using this research as a baseline, we can produce estimates for blood Pb levels of the ancient Roman specimens. Table S5 in the additional supporting information presents the calculated blood Pb levels for the Roman individuals based on the cumulative blood Pb index for the tibia (Flemming *et al.*, 1997). Our femur-based blood Pb calculations indicate that the majority of the Londinium Romans would have had blood Pb levels $> 5 \mu\text{g dl}^{-1}$. The highest blood Pb estimate for any bone was about $32 \mu\text{g dl}^{-1}$, the lowest was about $1.4 \mu\text{g dl}^{-1}$ and the average was $14.4 \mu\text{g dl}^{-1}$. Thus, our data suggest that clinically significant blood Pb levels reflecting chronic exposures were common.

With chronically elevated blood Pb levels, the health implications are known from medical and public health research of the past half century, but it has become recognized that individual variations in susceptibility occur and that there is no 'safe' level according to the CDC (2009). Children are more susceptible to neurotoxicity and anaemia due to a higher percentage of absorption of ingested Pb in the gut (Ziegler *et al.*, 1978), although adults may show the same disorders, along with hypertension and nephrotoxicity. In addition, an abundance of recent research has revealed that impaired reproductive capability unequivocally occurs in men, namely reduced sperm count and abnormal spermatozoa (Wirth and Mijal 2010) as Pb accumulates preferentially in testicles (Wang *et al.* 2013) and is toxic to sperm metabolic functions (He *et al.* 2016). Pregnant women may experience miscarriages, stillbirths or premature deliveries (Borja-Aburto *et al.*

1999). The effect of Pb exposure in reducing fertility has societal impact and could very well have contributed to population declines in Roman communities (Turchin and Scheidel, 2009). Given the growing body of evidence for Pb use and exposure in the Roman Empire, excessive Pb exposure and the potential effects on birth rate could have contributed to population decline in Londinium.

CONCLUSIONS

The data reveal that Londinium Roman bones contained 70-fold greater Pb levels than pre-Roman Iron Age individuals and show a distinct shift in bone Pb content that occurred during the Roman occupation of Britain. The Pb isotopic compositions in Romans are consistent with previously hypothesized widespread Pb pollution signals, while the Pb in pre-Romans of Britain was derived from distinct, localized regions. Because evidence of the Pb exposure was found in essentially all the Roman individuals studied, the vector would have been one or more, still unknown, ubiquitous practices among the Roman population, unrelated to age or sex, and which caused clinically elevated blood Pb levels in a significant portion (> 50%) of the Londinium population during the Roman occupation.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1 Strontium (Sr) and lead (Pb) purification procedure

Table S2a Strontium (Sr) isotopic compositions of standard reference materials measured during analyses for this study

Table S2b Lead (Pb) isotopic compositions of standard reference materials measured during analyses for this study

Table S3 Trace element analyses of Roman bone measured by High-resolution inductively coupled plasma mass spectrometry (HR-ICP-MS)

Table S4 Soil properties and lead (Pb) concentrations and isotopic compositions of soils measured by multi-collector inductively coupled plasma mass spectrometer (MC-ICP-MS)

Table S5 Blood lead (Pb) level estimates using cumulative blood Pb indices

Figure S1 Lead isotopediagrams comparing results for standard reference materials measured in this study to published data.

Figure S2 Box and whisker plot showing a lack of significant Pb concentration difference between male and female in Roman femora. The lack of difference between male and female specimens suggests that certain practices that were gender specific (e.g. use of cosmetics) were not the major sources of Pb exposure in Roman individuals.